THE SCHEME THEORETIC IMAGE OF A QUASI-COMPACT MORPHISM OF GROUPOIDS OVER A FIELD IS A GROUPOID SCHEME

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Allowed standard prerequisites (editorial).

- The morphism of groupoids over a field situation fixing R_1, R_2, a and the structure maps (more-groupoids chapter, Situation situation-morphism-groupoids-on-field, canonical tag 04Q5; kind reference). cited at statement lines 1849-1850. Supplies the notation and assumptions of the statement, here with a quasi-compact.
- Definition of the scheme theoretic image (Morphisms chapter, Definition morphisms-definition-scheme-theoretic-image, canonical tag 01R7). cited at statement lines 1852-1854. Fixes Z as the scheme theoretic image of a : R_1 to R_2.
- Scheme theoretic image of a product map equals the product of the scheme theoretic images (Varieties chapter, Lemma varieties-lemma-scheme-theoretic-image-product-map, canonical tag 04Q1). cited at proof lines 1870-1873. Identifies the scheme theoretic image of a x a with Z x_{s_2, U, t_2} Z, the key input showing the bottom arrow of the square maps it into Z.
- Groupoid scheme structures on images over fields: perfect field case with full proof of the identity and inverse restriction argument (more-groupoids chapter, Lemma lemma-map-groupoids-on-perfect-field-image, canonical tag 04Q8; collected on this lane as L22532). cited at proof lines 1875-1879. Its written proof supplies the verifications that i_2(Z) is contained in Z, that e_2 factors through Z, and the restriction of the groupoid axioms to Z, reused verbatim here.

Source reference. Original reference 04Q5: sources/stacks-chow/tags.txt, source lines 6138–6138. Cited at statement lines 1849-1850: notation and assumptions fixing R_1, R_2, a and the structure maps, with a quasi-compact.

Source reference. Original reference 01R7: sources/stacks-chow/tags.txt, source lines 2292–2292. Cited at statement lines 1852-1854: Z is the scheme theoretic image of a : R_1 to R_2.

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Source reference. Original lemma 04Q1: sources/stacks-chow/tags.txt, source lines 6134–6134. Cited at proof lines 1870-1873: the scheme theoretic image of a x a equals Z x_{s_2, U, t_2} Z, whence the bottom arrow of the commutative square maps it into Z.

Source reference. Original lemma 04Q8: sources/stacks-chow/tags.txt, source lines 6141–6141. Cited at proof lines 1875-1879: its written proof supplies the i_2 and e_2 restriction verifications and the groupoid axiom restriction, reused as in that proof.

Lemma 0.2. *Notation and assumptions as in Situation 04Q5. Assume that $a : R_1 \to R_2$ is a quasi-compact morphism. Let $Z \subset R_2$ be the scheme theoretic image (see Morphisms, Definition 01R7) of $a : R_1 \to R_2$. Then*

$$(U, Z, s_2|_Z, t_2|_Z, c_2|_Z)$$

is a groupoid scheme over S.

Proof. The main difficulty is to show that $c_2|_{Z \times_{s_2,U,t_2} Z}$ maps into Z. Consider the commutative diagram

$$\begin{array}{ccc} R_1 \times_{s_1,U,t_1} R_1 & \longrightarrow & R_1 \\ \downarrow{\scriptstyle a \times a} & & \downarrow \\ R_2 \times_{s_2,U,t_2} R_2 & \longrightarrow & R_2 \end{array}$$

By Varieties, Lemma 04Q1 we see that the scheme theoretic image of $a \times a$ is $Z \times_{s_2,U,t_2} Z$. By the commutativity of the diagram we conclude that $Z \times_{s_2,U,t_2} Z$ maps into Z by the bottom horizontal arrow. As in the proof of Lemma 04Q8 it is also true that $i_2(Z) \subset Z$ and that e_2 factors through Z. Hence we conclude as in the proof of that lemma. □